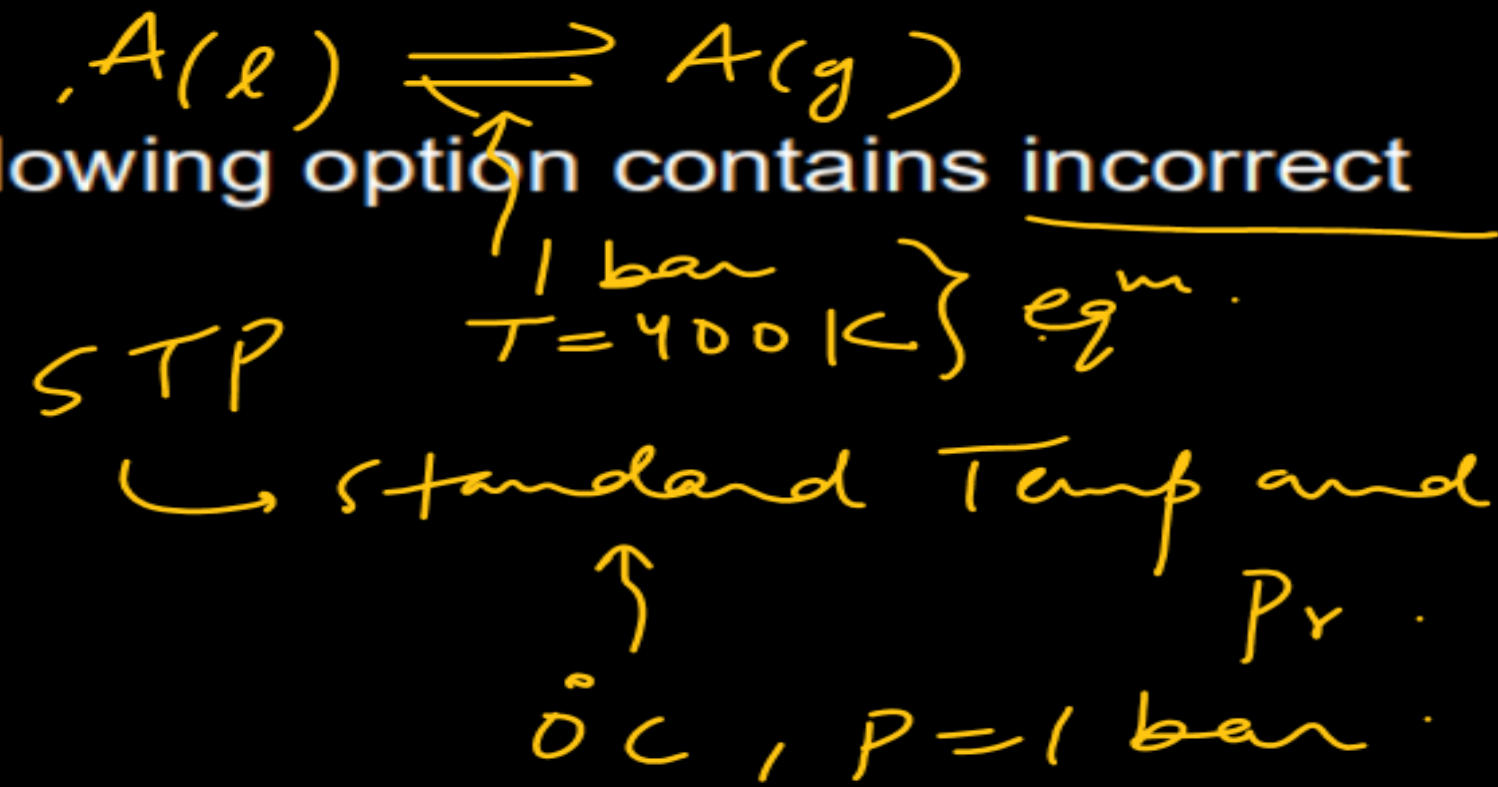


✓ Good

A substance A has a standard boiling point of 400 K. Which of the following option contains incorrect set of thermodynamic parameters for the following change.

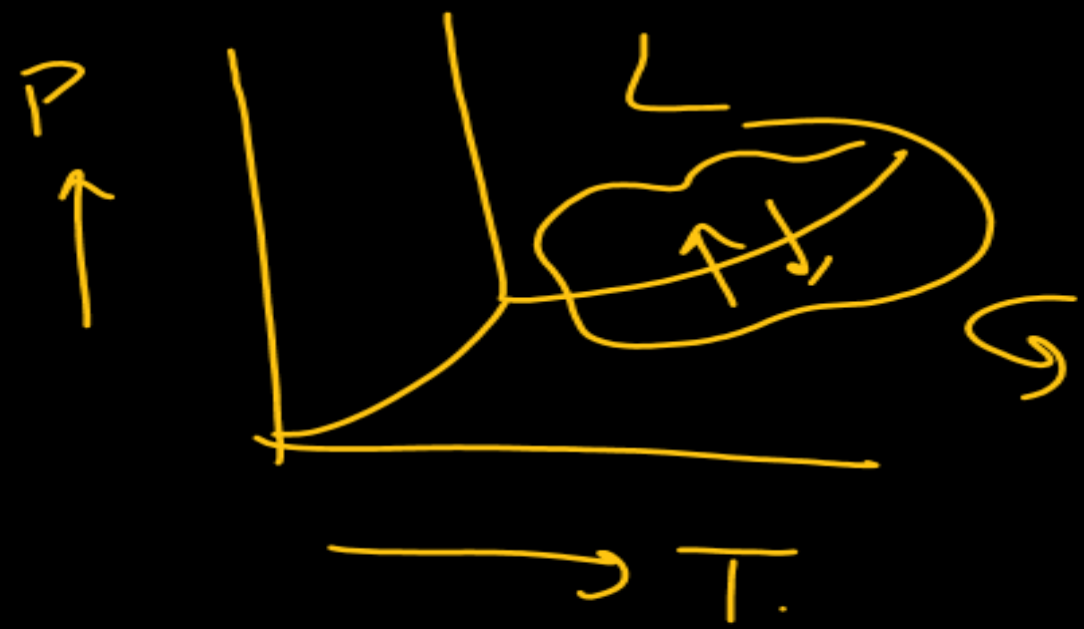


non-spontaneous

- (A) If $P_1 = 1 \text{ atm}$ and $T_1 = 350 \text{ K}$ then $\Delta G > 0, q > 0, W < 0$ ✓
- (B) If $P_1 = 2 \text{ bar}$ and $T_1 = 400 \text{ K}$ then $\Delta G > 0, w < 0, \Delta H > 0$ ✓
- ✓ (C) If $P_1 = 1 \text{ atm}$ and $T_1 = 400 \text{ K}$ then $\Delta G = 0, w < 0, \Delta S > 0$ ✓
- (D) If $P_1 = \frac{1}{2} \text{ atm}$ and $T_1 = 450 \text{ K}$ then $\Delta G < 0, \Delta S_{\text{Surroundings}} < 0, q > 0$ ✓

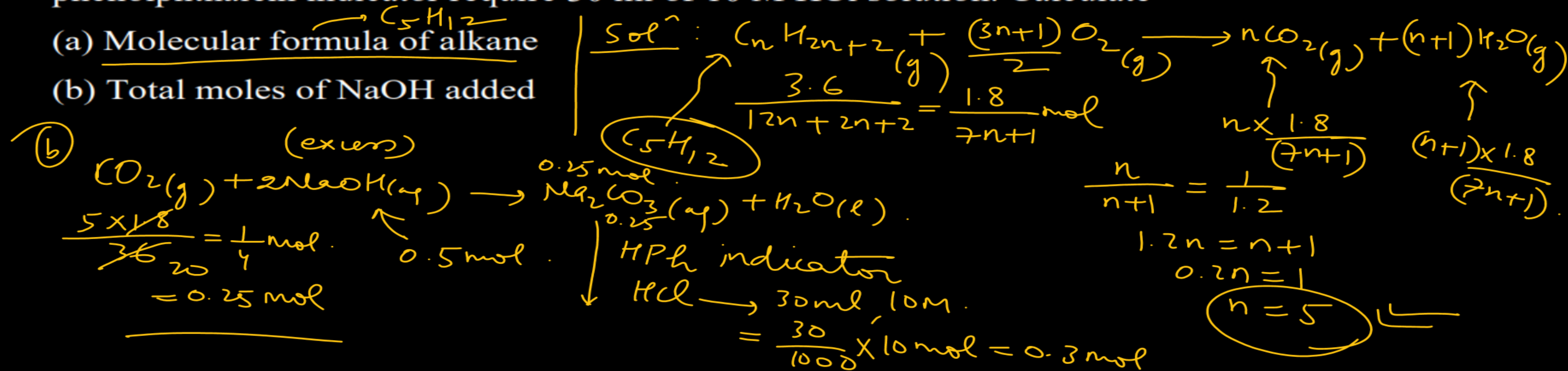
NBP of water = 100°C
 $\rightarrow P = 1 \text{ atm}$

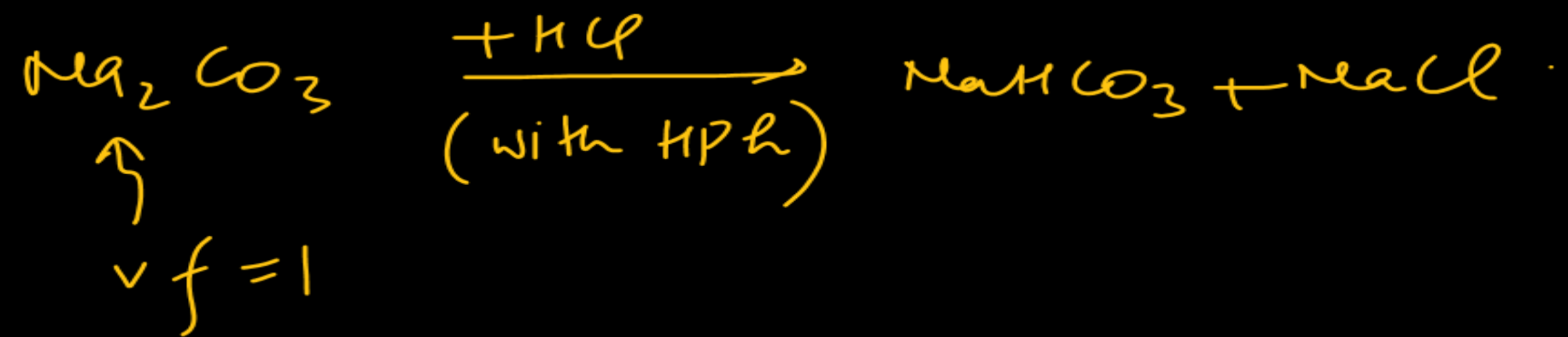
SBP $\rightarrow 99.6^\circ\text{C}$
 $\rightarrow P_r = 1 \text{ bar}$
 standard B.P.
 $P \uparrow, \text{BP} \uparrow$



3.6 g of an alkane on complete combustion at 400 K gave CO_2 & H_2O gas whose volumes at similar temperature & pressure were found to be in the ratio 1 : 1.2. If all the CO_2 produced was reacted with excess of NaOH solution & the resulting solution in presence of phenolphthalein indicator require 30 ml of 10 M HCl solution. Calculate

- (a) Molecular formula of alkane
 (b) Total moles of NaOH added





$$0.25 \times 1 = n_{\text{HCl}} \times 1$$

$$\Rightarrow n_{\text{HCl}} = 0.25 \text{ mol}$$

$$(n_{\text{HCl}})_{\text{used for excess of NaOH}} = 0.3 - 0.25 \\ = 0.05 \text{ mol}$$

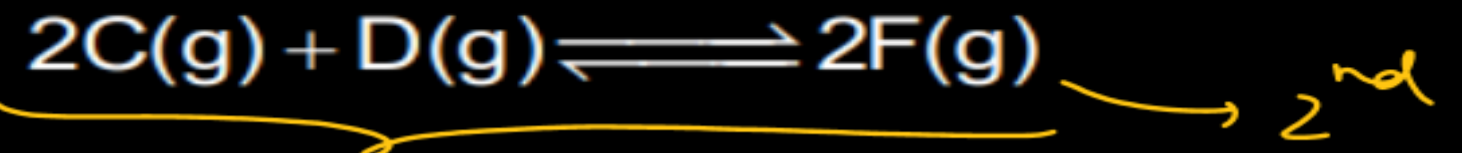
$$\text{excess mol. of NaOH} = 0.05 \text{ mol}$$

$$(n_{\text{T}})_{\text{NaOH}} = 0.5 \text{ mol} + 0.05 \text{ mol} \\ = \underline{0.55 \text{ mol}} \quad \text{Ans}$$

✓ Good

Simultaneous equilibrium.

Three ideal gases A, B & D were taken in a vessel of constant volume in molar ratio of 1 : 2 : 3 respectively causing the reaction : $A(g) + 2B(g) \rightleftharpoons C(g)$. After a long time when the equilibrium (1st) is established the total pressure was $\frac{5}{6}$ th of the initial total pressure. At this point volume was reduced to half and a catalyst was added starting the reaction.



If at equilibrium (2nd) moles of A & C are equal then find the ratio of equilibrium total pressure at 2nd equilibrium to equilibrium total pressure at 1st equilibrium.

Solⁿ:

$A + 2B \rightleftharpoons C$

Initial: $p \quad 2p \quad 0$

Change: $-x \quad -2x \quad +x$

Equilibrium: $p-x \quad 2p-2x \quad x$

$K_p = \frac{p}{2} \times \frac{p^2}{p/2 \times p^2}$

$K_p = \frac{1}{p^2}$

$(p_T)_f = (p-x) + 2p-2x + x = \frac{5}{6} \times (p + 2p + 3p)$

$6p - 2x = 5p \Rightarrow x = p/2$

A, B, D

↓ ↓ ↓

$p \quad 2p \quad 3p$

Ans : 1.7



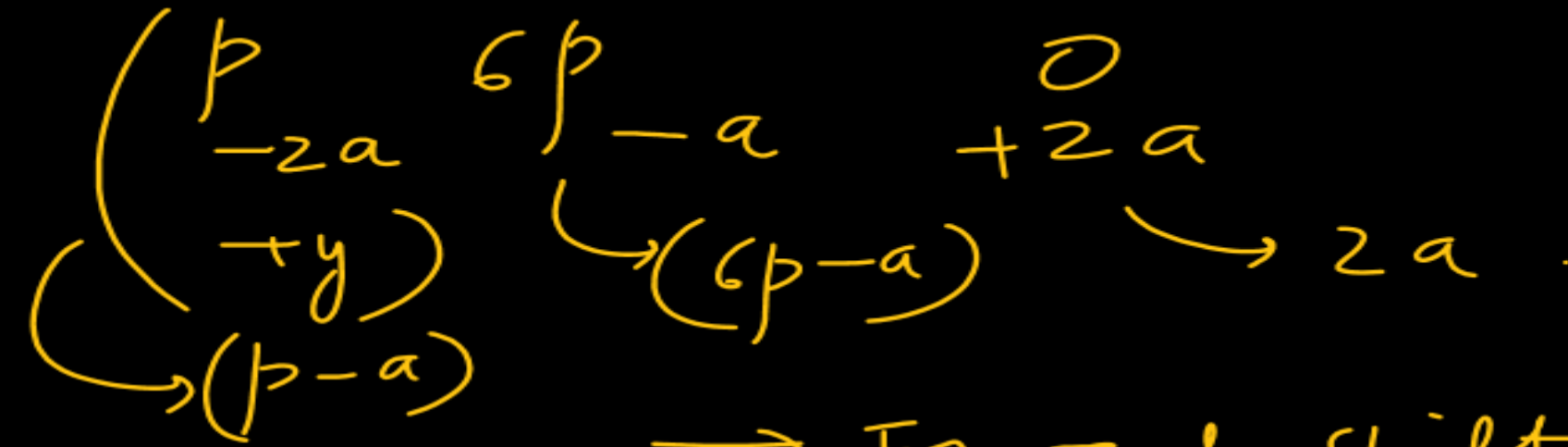
$$p_A = p_C$$

at new eq.

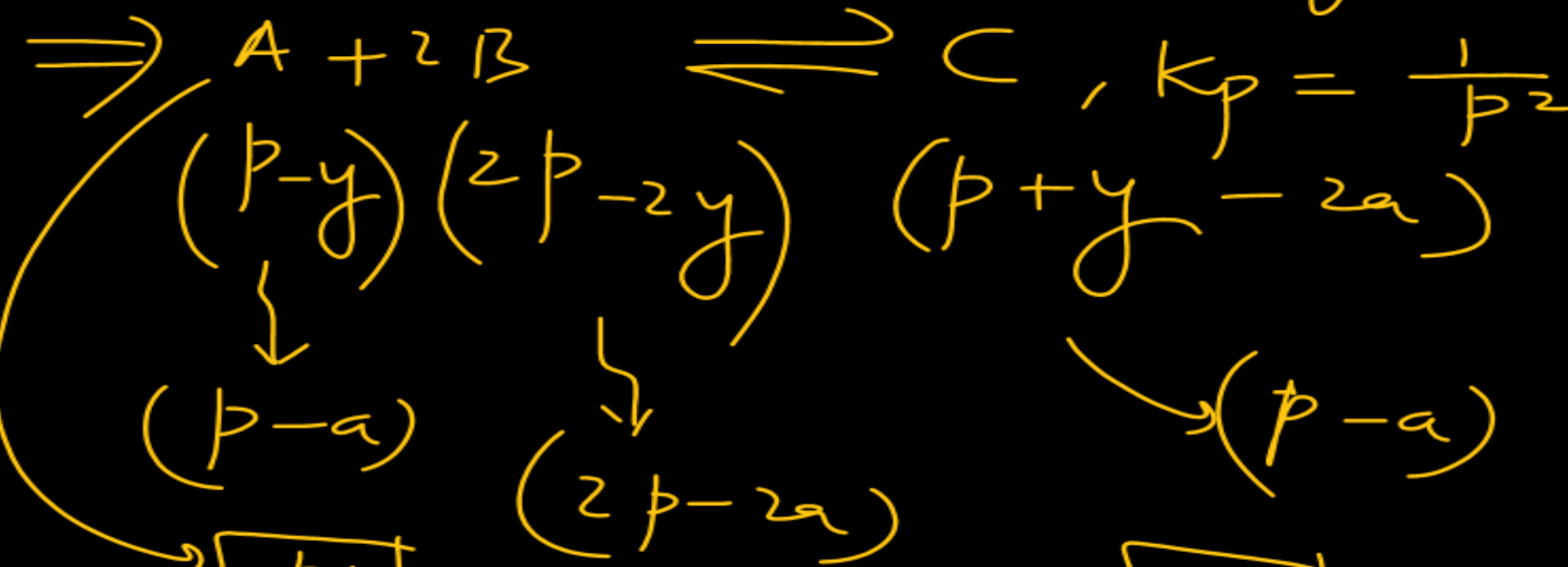
$$p - y = p + y - 2a$$

$$2a = 2y$$

$$a = y$$



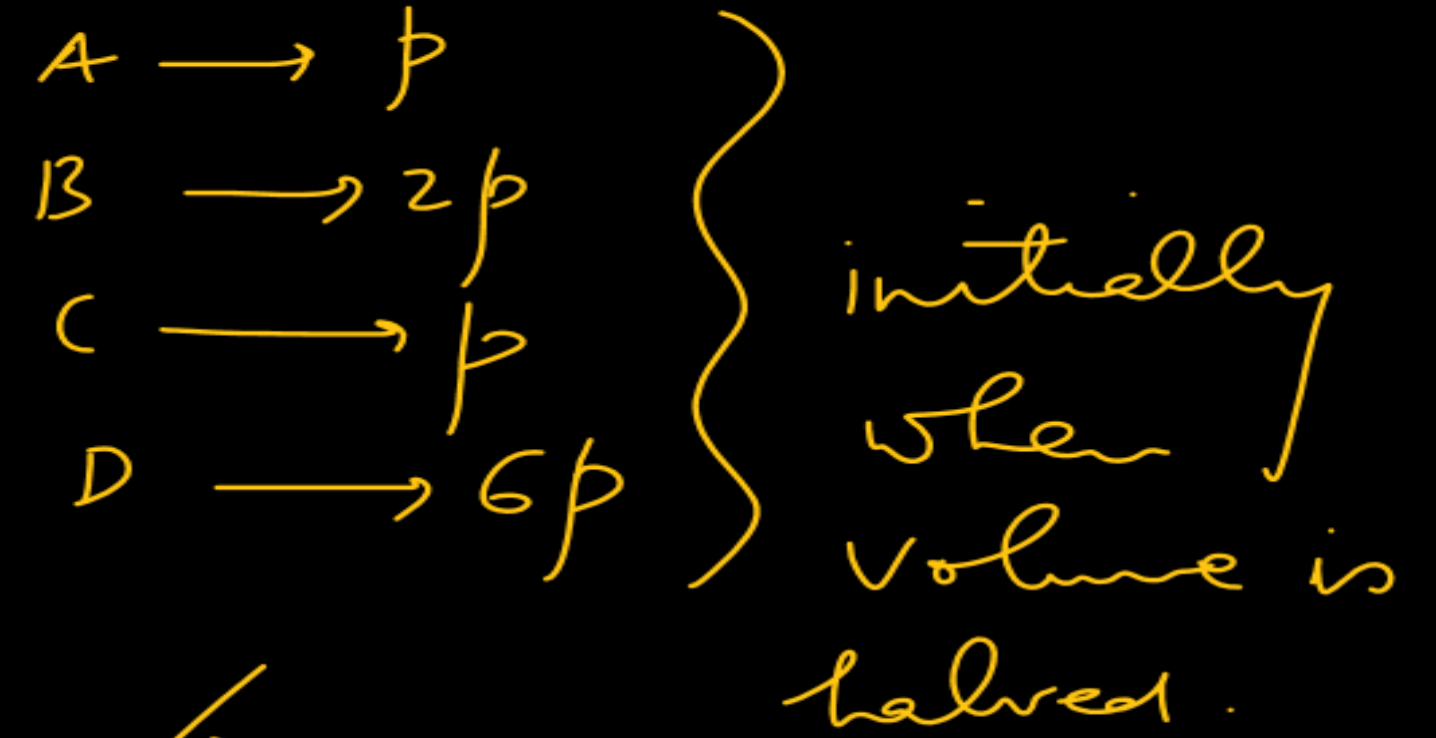
Forward shift



$$p^2 = (2p-2a)^2$$

$$2p-2a = p$$

$$a = p/2$$

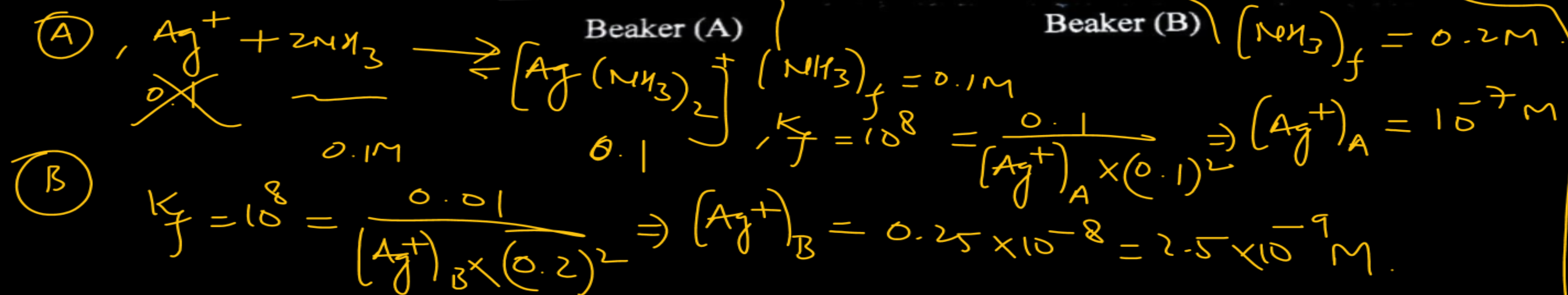
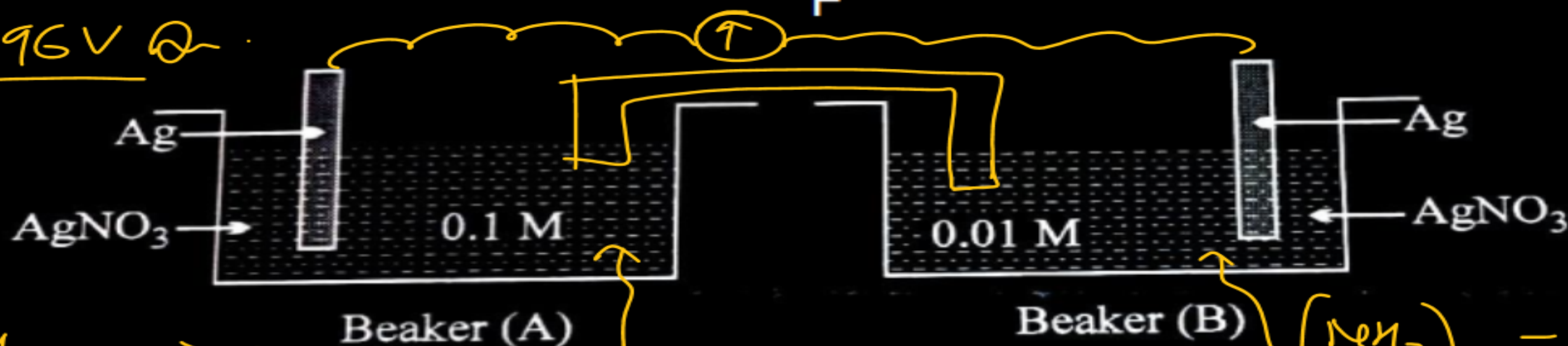


$$\frac{(p_T)_{eq, now}}{(p_T)_{eq, earlier}} = \frac{p/2 + p + p/2 + 5.5p + p}{5p} = \frac{8.5p}{5p} = 1.7 \text{ atm}$$

Two beakers A and B contains AgNO_3 solutions of concentrations 0.1 M and 0.01 M respectively as shown in figures. Pure silver electrodes are now lowered in each beaker. Now NH_3 is added in each beaker till conc. Of NH_3 in beaker A and B become 0.1 M and 0.2 M respectively. If two beakers are now connected in Galvanic cell arrangement, find the EMF (in Volts) of the resulting cell at 298 K.

Given : $E^\circ_{\text{Ag}^+/\text{Ag}} = 0.8 \text{ Volt}$, $K_f[\text{Ag}(\text{NH}_3)_2^+] = 10^8$, $\frac{RT}{F}(2.303) = 0.06$: [use $\log_{10} 2 = 0.3$]

Ans: $E_{\text{cell}} = 0.096 \text{ V}$



$\text{Ag}^+_{(B)} \longrightarrow \text{Ag}^+_{(A)}$

$$E = 0 - \frac{0.06}{1} \log \frac{10^{-7}}{2.5 \times 10^{-9}}$$

$$= -0.06 \log [40]$$

$$= -0.06 [1 + 0.6]$$

$$= -0.096 \text{ volt}$$

cell works as

$\text{Ag}^+_{(A)} \longrightarrow \text{Ag}^+_{(B)}$

$E_{\text{cell}} = 0.096 \text{ V}$

A 0.1 M solution of $[\text{Cu}(\text{NH}_3)_4]^+$ is stirred with an excess of potassium cyanide sufficient to convert all the ammonium complex to the corresponding cuprocyanide complex $[\text{Cu}(\text{CN})_4]^{3-}$ and in addition to provide the solution with an excess of CN^- equal to 0.2 M. Calculate the maximum pH of the solution when the final solution is treated with hydrogen sulphide to maintain $[\text{H}_2\text{S}] = 0.1 \text{ M}$ and the precipitation of cuprous sulphide is prevented.

$\text{pH} = 10$

The instability constant for $[\text{Cu}(\text{CN})_4]^{3-}$ is 6.4×10^{-15} , $K_{a, \text{overall}}$ of $\text{H}_2\text{S} = 1.6 \times 10^{-21}$,

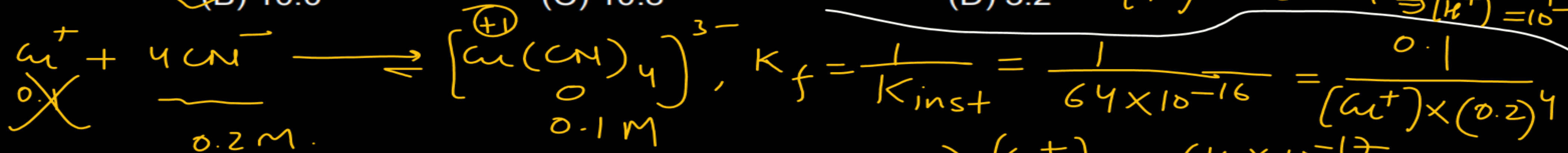
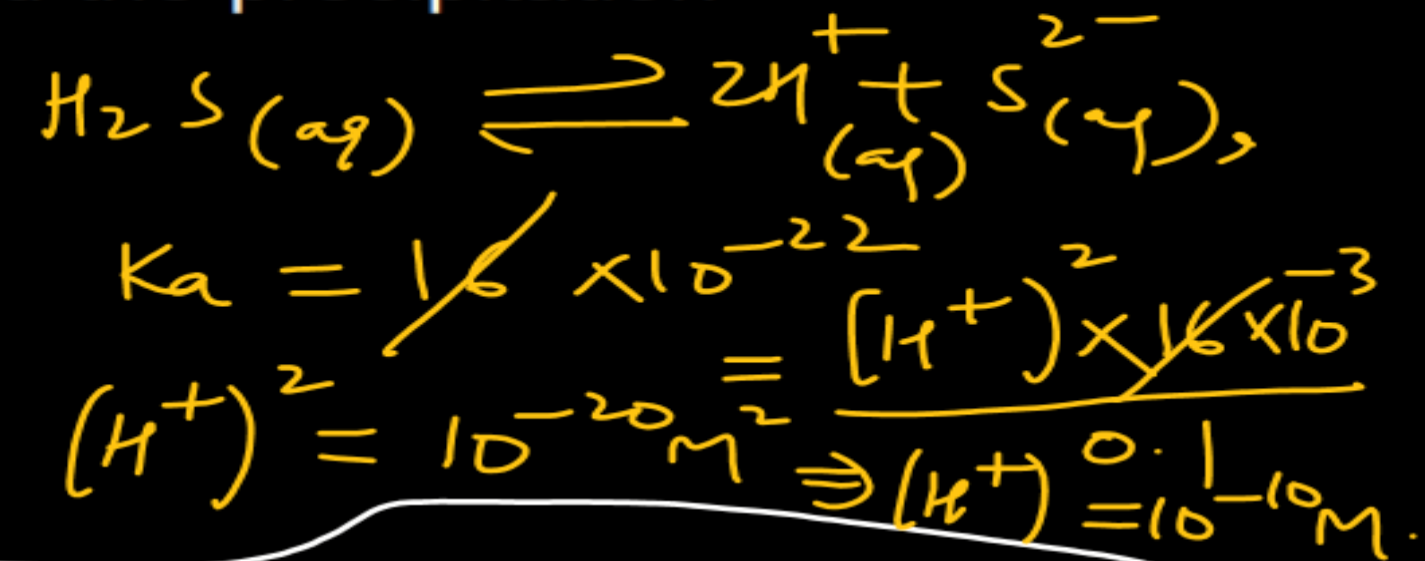
K_{sp} of $\text{Cu}_2\text{S} = 2.56 \times 10^{-27}$.

(A) 4.0

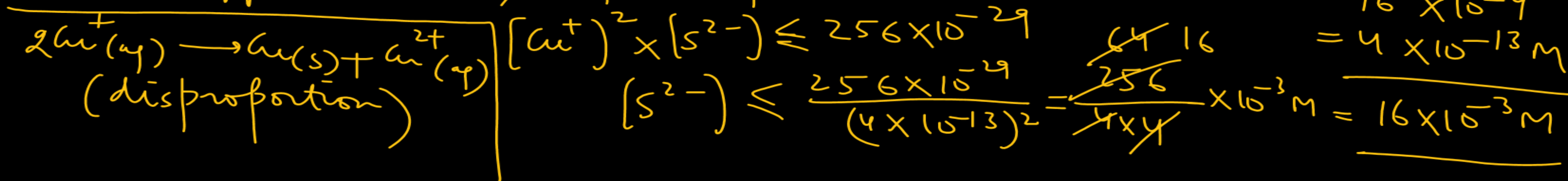
(B) 10.0

(C) 10.8

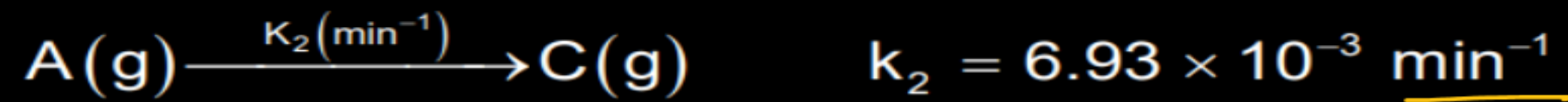
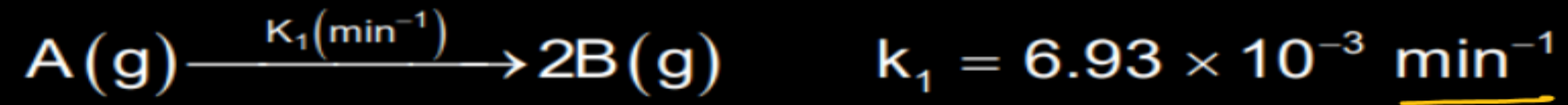
(D) 3.2



No pptⁿ means, $Q_{ip} \leq K_{sp}$



A compound 'A' transform by two parallel first order paths



$$(p_T)_{at eq} = p_A + p_B + p_C$$

$$= 1 + 1 + \frac{1}{2}$$

$$= \underline{2.5 \text{ atm}} \quad \underline{\text{Ans.}}$$

If reaction started with pure 'A' with conc. 0.1 mol/L in a closed container with initial pressure 2 atm.

What is the pressure (in atm) developed in container after 50 minutes from start of experiment.

Solⁿ: 0.1 M A $\begin{cases} \xrightarrow{k_1} 2B \\ \xrightarrow{k_2} C \end{cases}$ $y \text{ atm}$

$t=0, 2 \text{ atm}$

$t=50 \text{ min}, (2-x)$

$2-x = 1 \text{ atm}$

$\Rightarrow x = 1 \text{ atm}$

$x \begin{cases} \rightarrow y/2 \\ \rightarrow z \end{cases} \Rightarrow x = \frac{y}{2} + z \quad \text{--- (1)}$

$\frac{(B)_t}{(C)_t} = \frac{2k_1}{1k_2} = 2 = \frac{y}{z} \Rightarrow y = 2z$

$x = \frac{y}{2} + z \Rightarrow 1 = z + z \Rightarrow \underline{z = \frac{1}{2} \text{ atm}}$

$y = 2 \times \frac{1}{2} = 1 \text{ atm}$

$(A)_t = (A)_0 e^{-(k_1+k_2) \times t}$

$(p_A)_{50 \text{ min}} = 2 \times e^{-2 \times 6.93 \times 10^{-3} \times 50}$

$= 2 \times e^{-0.693}$

$= 2 \times e^{-\ln 2}$

$= 1 \text{ atm}$

Ans: 4.33 g/cc @

CCP

The density of a pure substance 'A' whose atoms pack in cubic close pack arrangement is 1 g/cc. If B atoms can occupy tetrahedral void and if all the tetrahedral voids are occupied by 'B' atom. What is the density of resulting solid in g/cc. [Atomic mass (A) = 30 g/mol & atomic mass (B) = 50 g/mol]

$$d_A = 1 \text{ g/ml} = \frac{4 \times \frac{M_A}{N_A}}{a^3}, \quad a = \text{edge length of cube unit cell.}$$

$A_4 B_8 \Rightarrow$ AB_2 Formula.

$N_A \rightarrow$ Avogadro's No.

$$d_f = \frac{4 \times (M_A + 2 \times M_B)}{N_A \times a^3}$$
$$\frac{d_A}{d_f} = \frac{1}{d_f} = \frac{M_A}{(M_A + 2 \times M_B)} \Rightarrow \frac{1}{d_f} = \frac{30}{30 + 2 \times 50} \Rightarrow d_f = \frac{130}{30} @$$
$$= \underline{4.33 \text{ g/cc}}$$